THE PLEURA

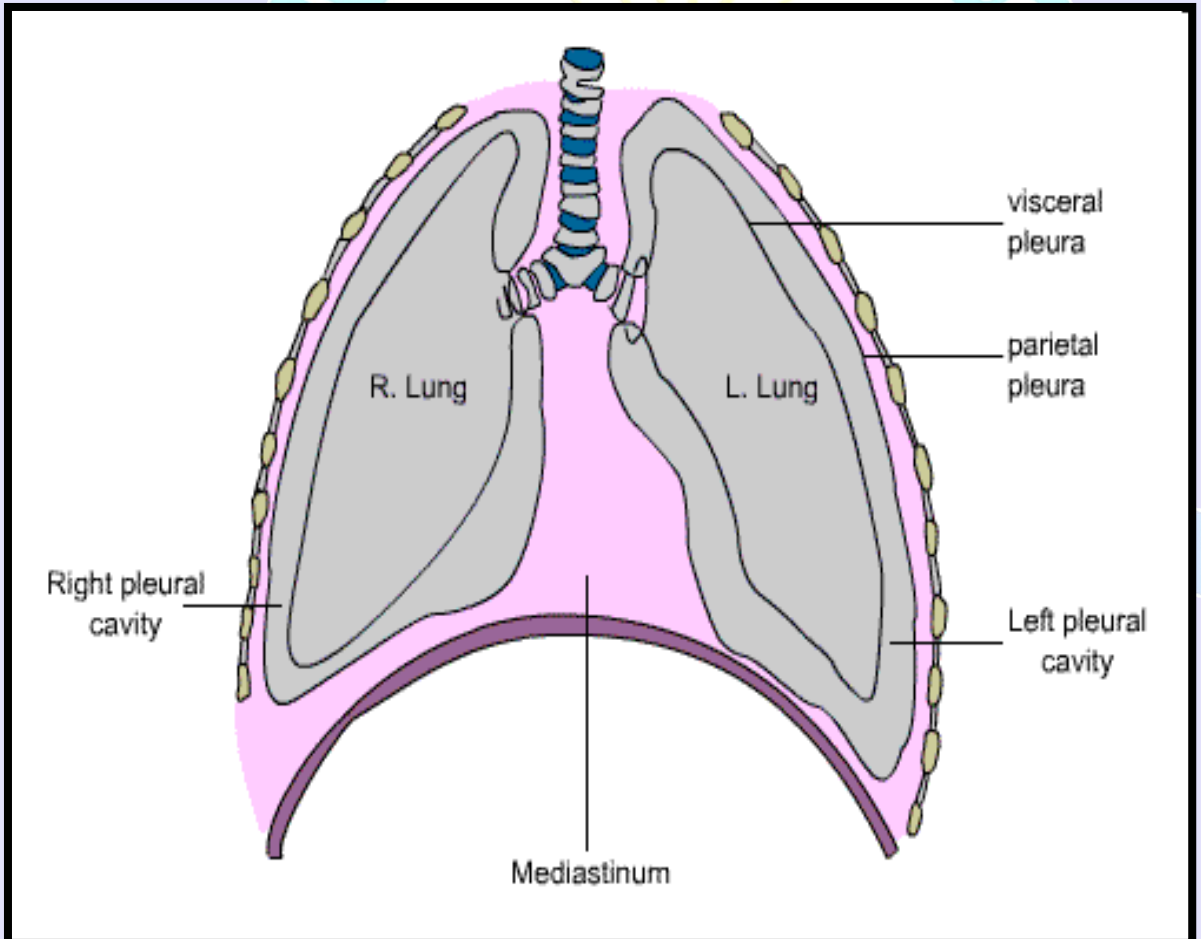
LEARNING OBJECTIVES

By the end of the lecture the students should be able to identify:

- The gross features of pleura
- The pleural cavity and the pleural reflections
- The surface anatomy related to pleural reflections
- The clinical application related to the topic.

THE PLEURA

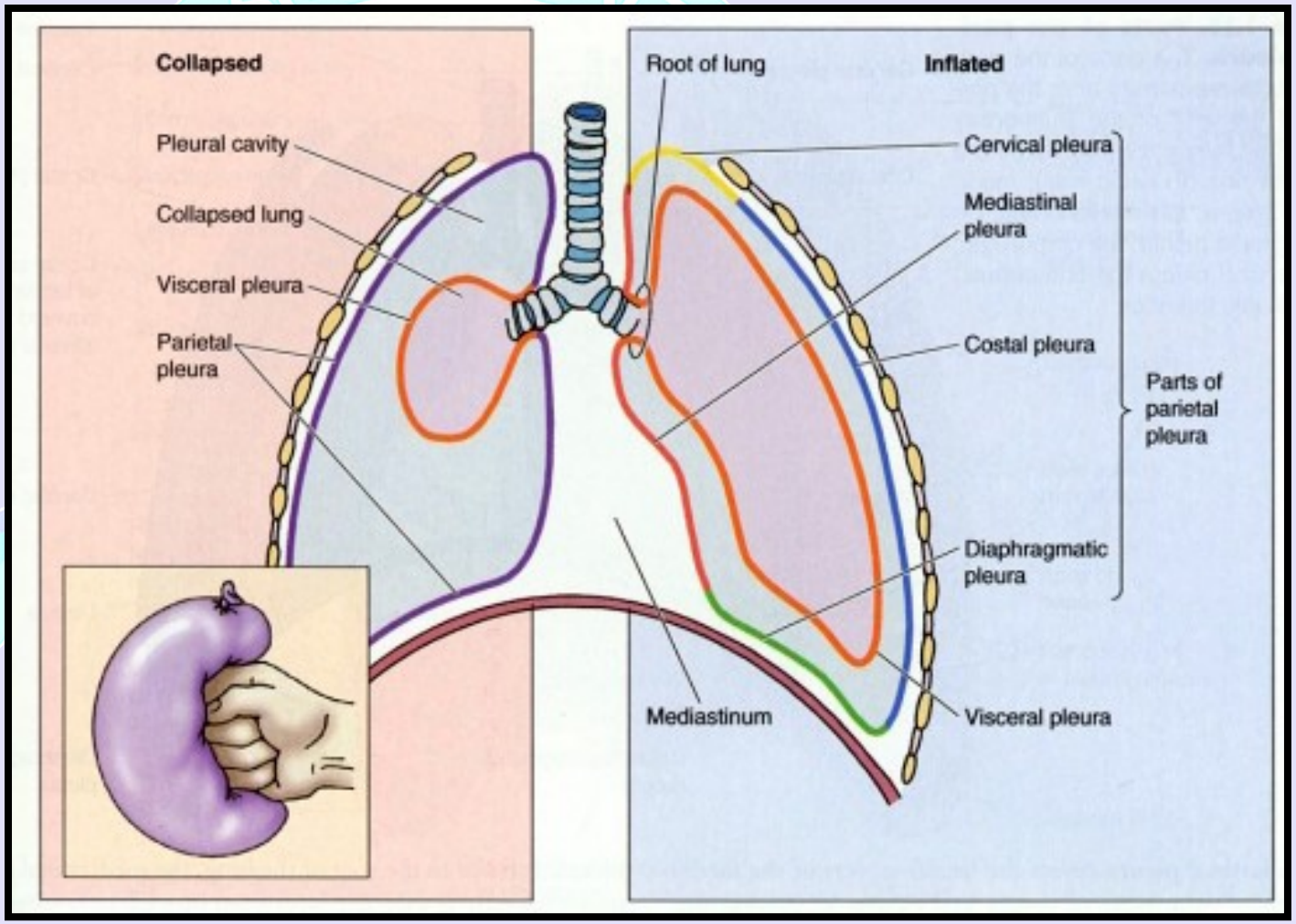
Double layered serous membrane that invests the lungs and lines the thoracic cavity.



THE PLEURA

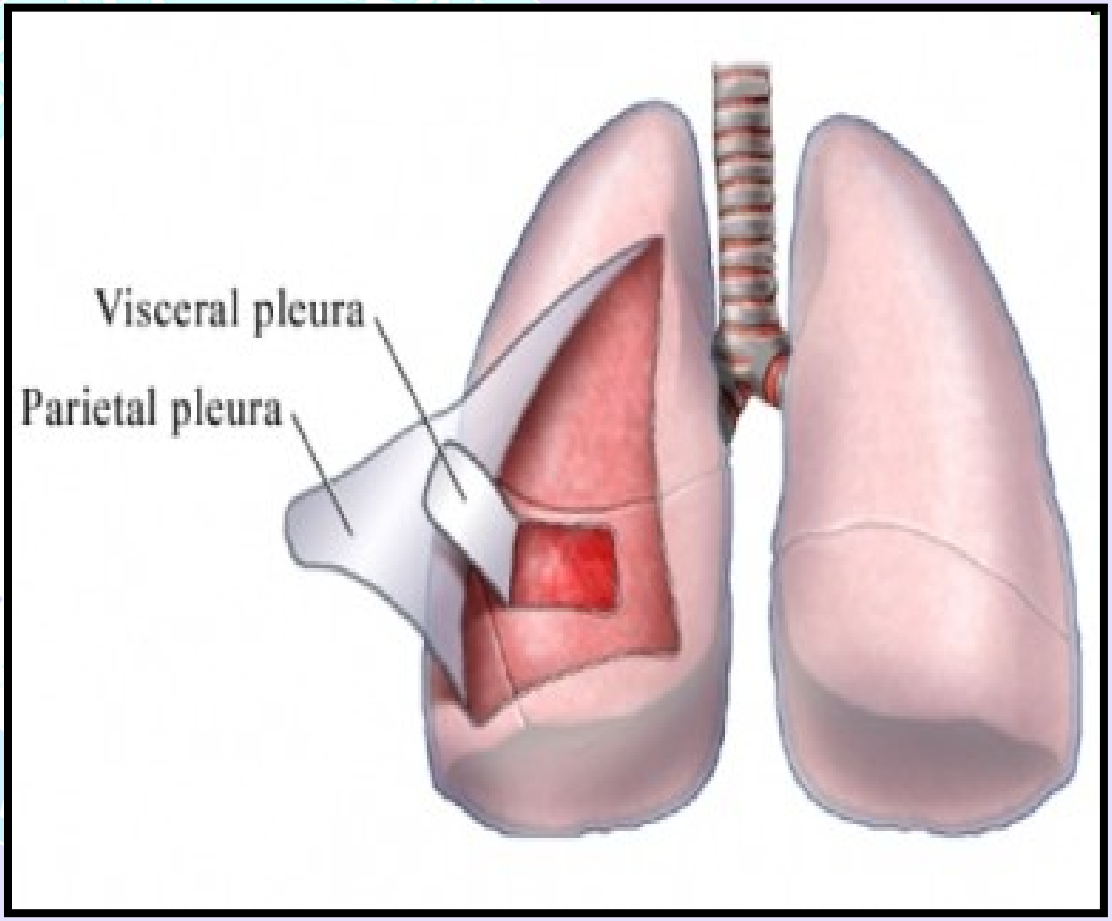
Two layers:

- Visceral pleura
- Parietal pleura



VISCERAL PLEURA

- Layer of pleura lining the surface of the lungs
- Thin and adherent to lung surface
- Dips in all the fissures.
- Insensitive.

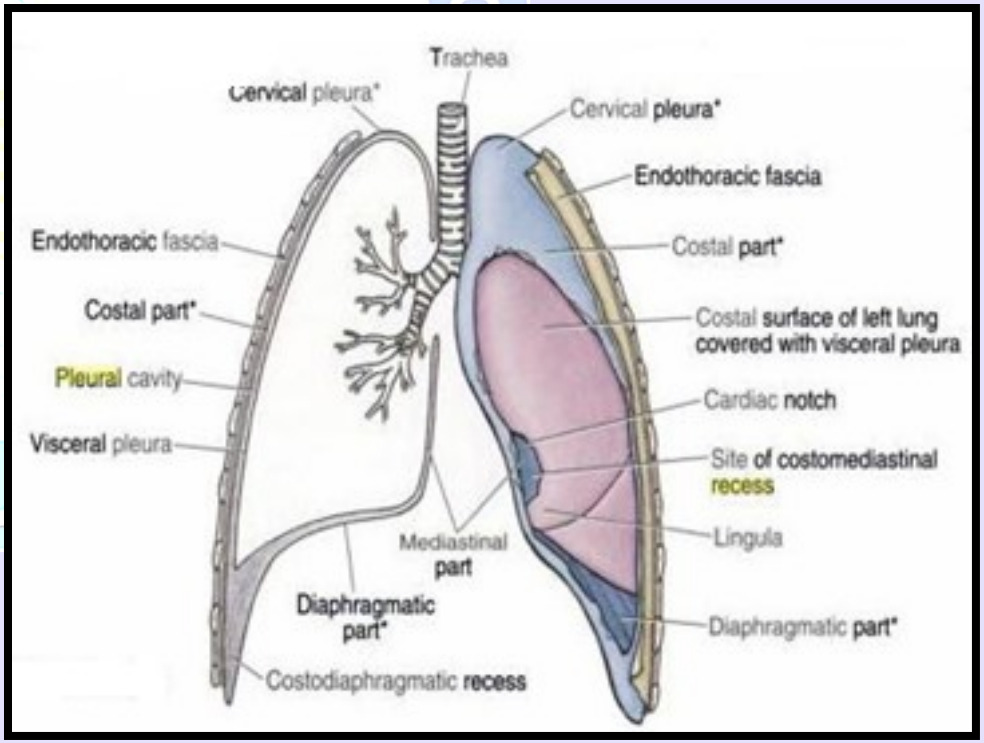


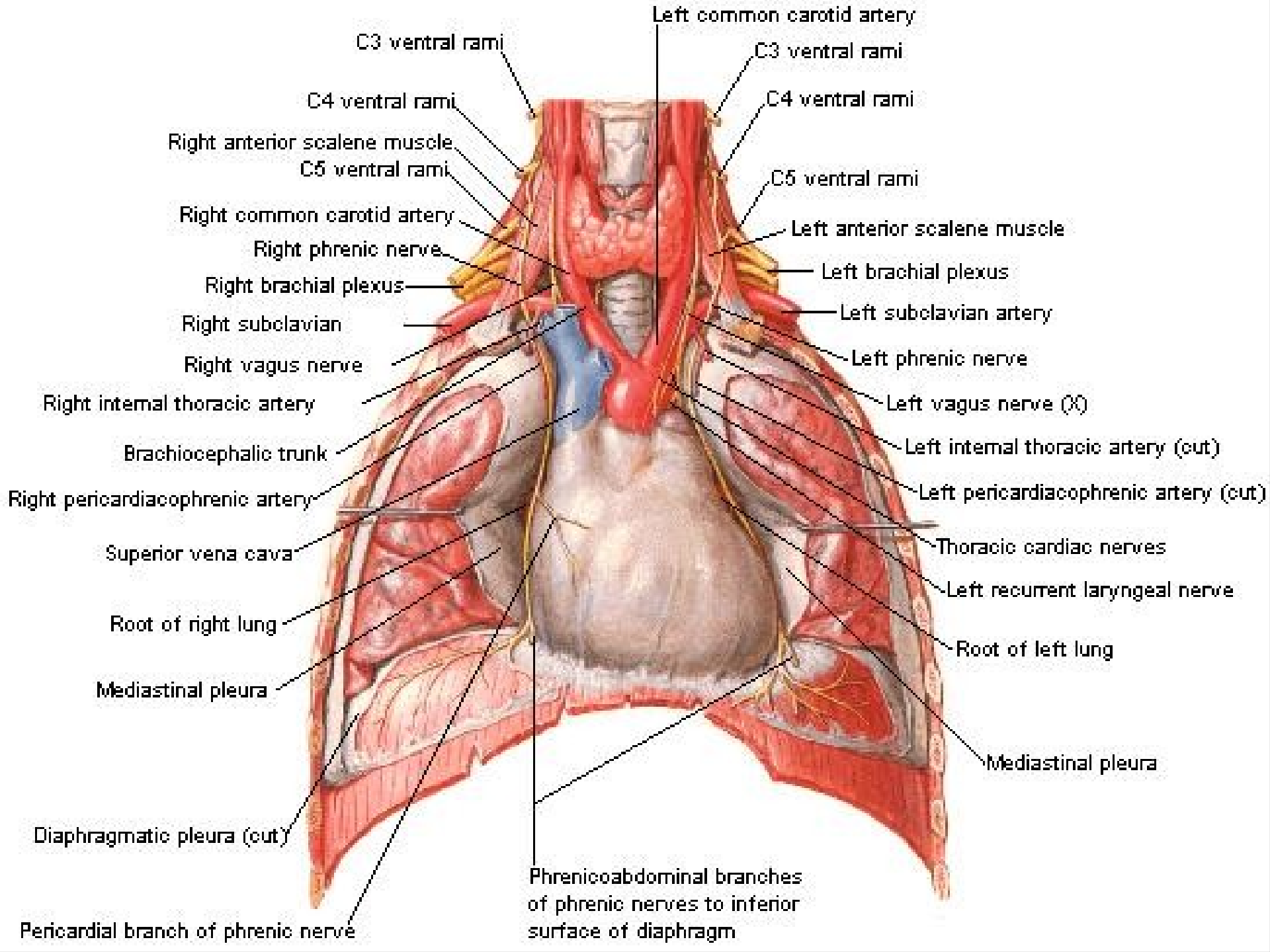
PARIETAL PLEURA

- **Lines the thoracic cavity.**
- **Sensitive layer**

➢ **Parietal pleura is divided into the region in which it lies or the surface that it covers**

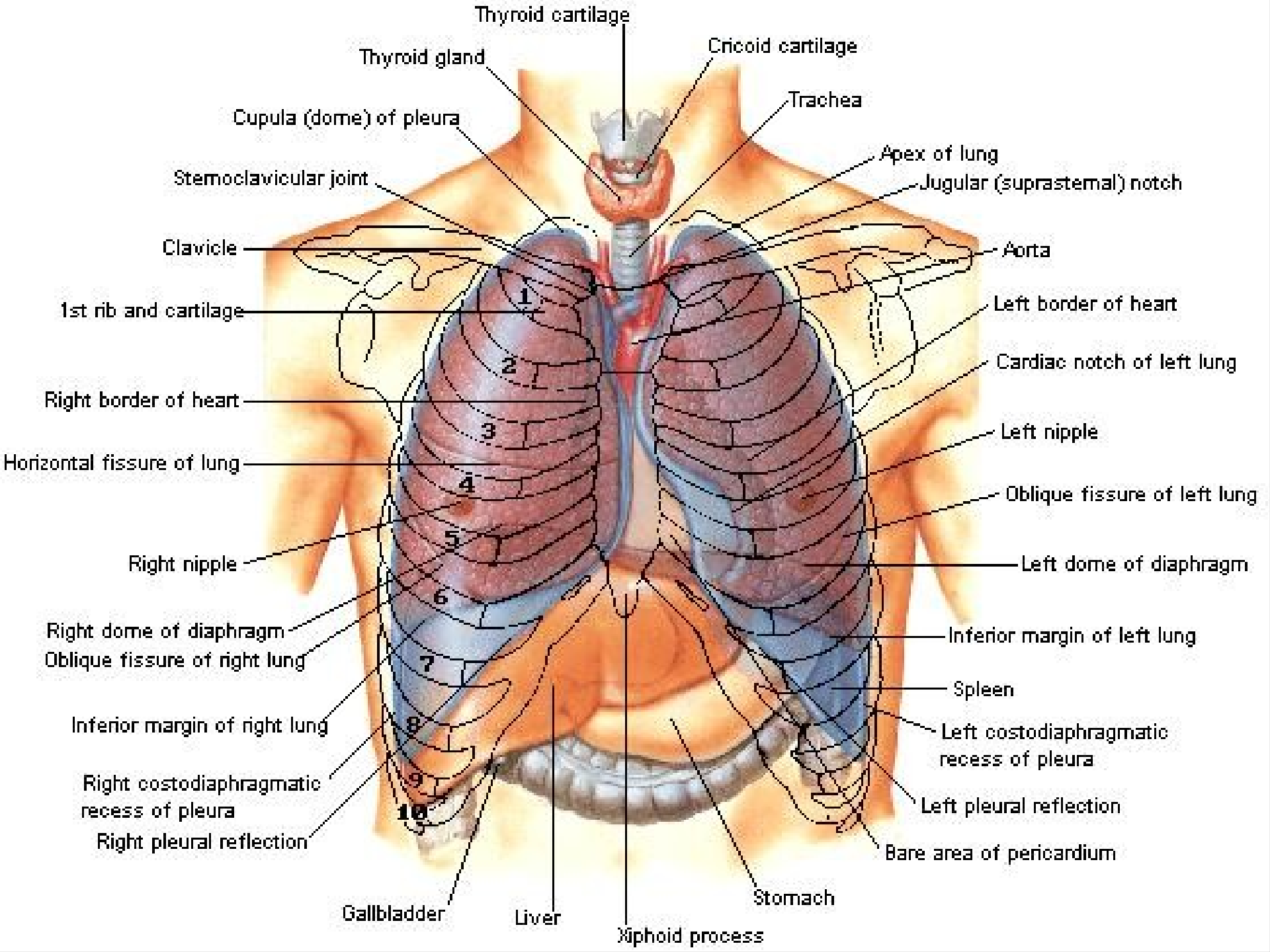
- Cervical
- Costal
- Diaphragmatic
- Mediastinal




Left common carotid artery
C3 ventral rami
C3 ventral rami
C4 ventral rami
C4 ventral rami
Right anterior scalene muscle
C5 ventral rami
C5 ventral rami
Right common carotid artery
Left anterior scalene muscle
Right phrenic nerve
Left brachial plexus
Right brachial plexus
Left subclavian artery
Right subclavian
Left phrenic nerve
Right vagus nerve
Left vagus nerve (X)
Right internal thoracic artery
Left internal thoracic artery (cut)
Brachiocephalic trunk
Left pericardiacophrenic artery (cut)
Right pericardiacophrenic artery
Thoracic cardiac nerves
Superior vena cava
Left recurrent laryngeal nerve
Root of right lung
Root of left lung
Mediastinal pleura
Mediastinal pleura
Diaphragmatic pleura (cut)
Phrenicoabdominal branches
of phrenic nerves to inferior
surface of diaphragm
Pericardial branch of phrenic nerve

CERVICAL PLEURA

- The cervical pleura extends up into the neck
- It lines the undersurface of the suprapleural membrane
- It reaches a level 1 to 1.5 in. (2.5 to 4 cm) above the medial third of the clavicle


Thyroid cartilage
Thyroid gland
Cricoid cartilage
Trachea
Cupula (dome) of pleura
Apex of lung
Sternoclavicular joint
Jugular (suprasternal) notch
Clavicle
Aorta
1st rib and cartilage
Left border of heart
Cardiac notch of left lung
Right border of heart
Left nipple
Horizontal fissure of lung
Oblique fissure of left lung
Right nipple
Left dome of diaphragm
Right dome of diaphragm
Inferior margin of left lung
Oblique fissure of right lung
Spleen
Inferior margin of right lung
Left costodiaphragmatic recess of pleura
Right costodiaphragmatic recess of pleura
Left pleural reflection
Right pleural reflection
Bare area of pericardium
Stomach
Gallbladder
Liver
Xiphoid process
1
2
3
4
5
6
7
8
9
10

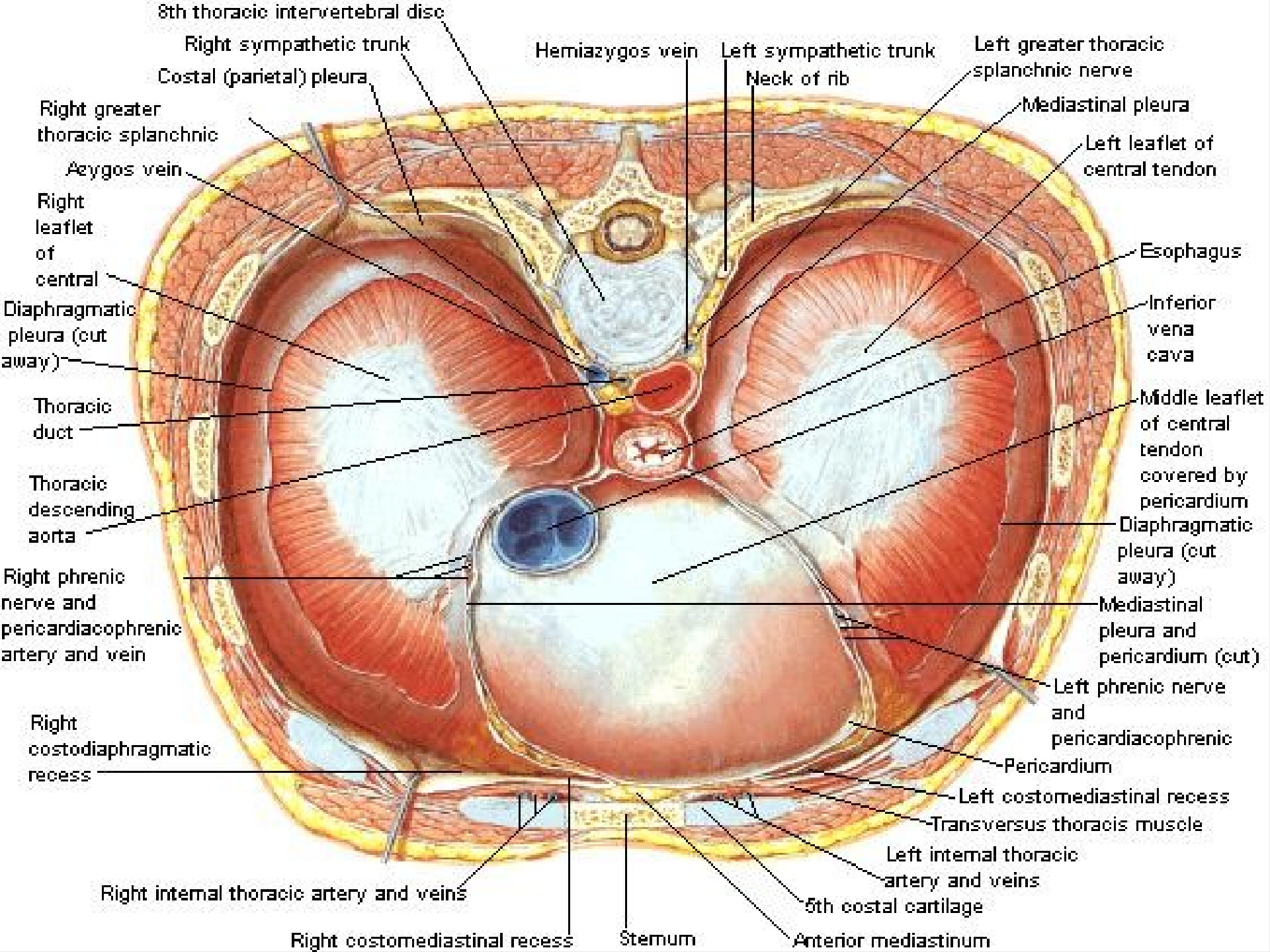
COSTAL PLEURA

- It lines the inner surfaces of:

  - The ribs
  - The costal cartilages
  - The intercostal spaces
  - The sides of the vertebral bodies
  - The back of the sternum

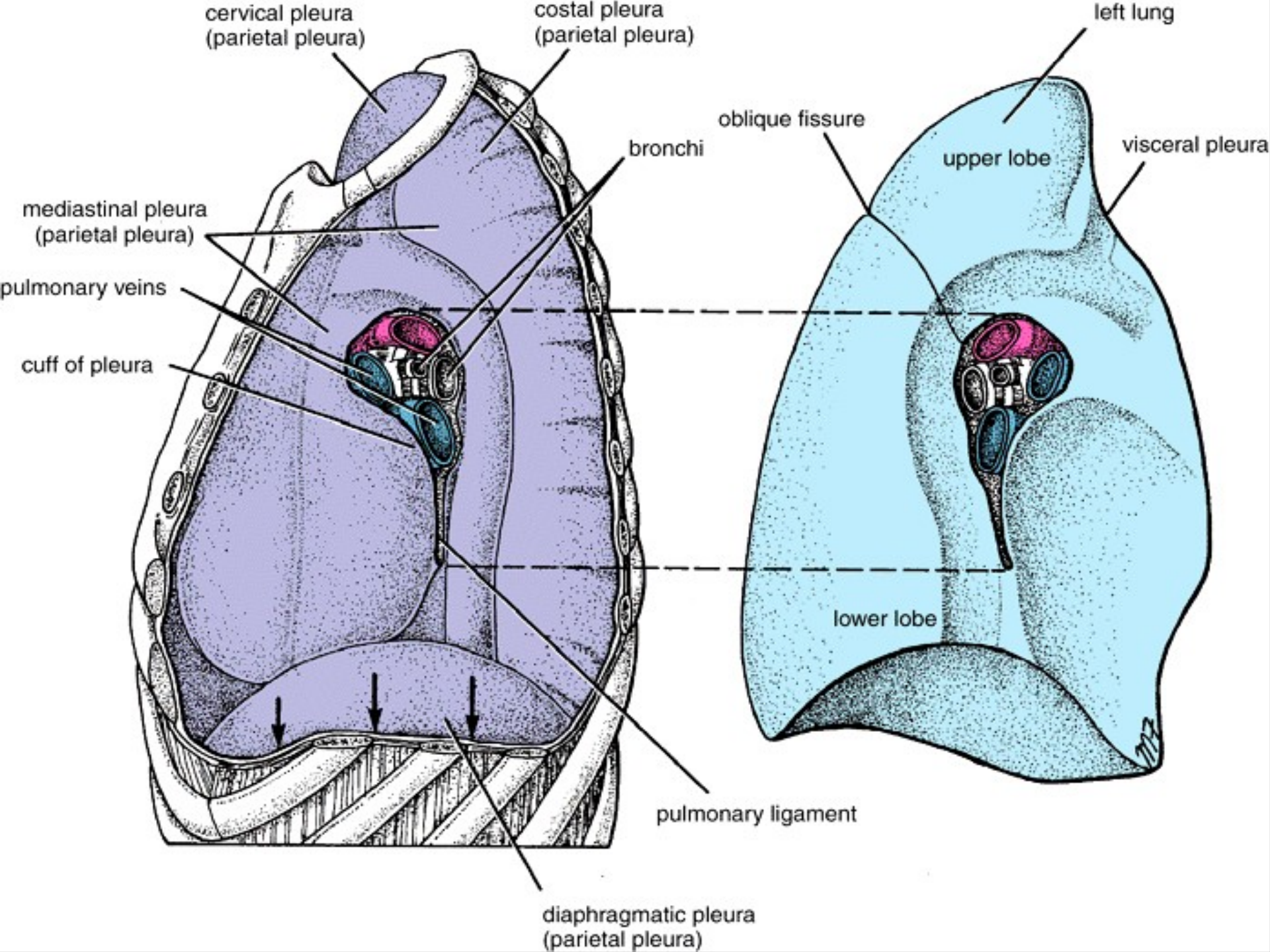
DIAPHRAGMATIC PLEURA

- It covers the thoracic surface of the diaphragm
- covers the surface of the diaphragm except for the middle part, occupied by the pericardium, and a narrow strip at the costal attachment
- In quiet respiration, the costal and diaphragmatic pleurae are in apposition to each other below the lower border of the lung
- Costal and diaphragmatic pleurae separate in deep inspiration

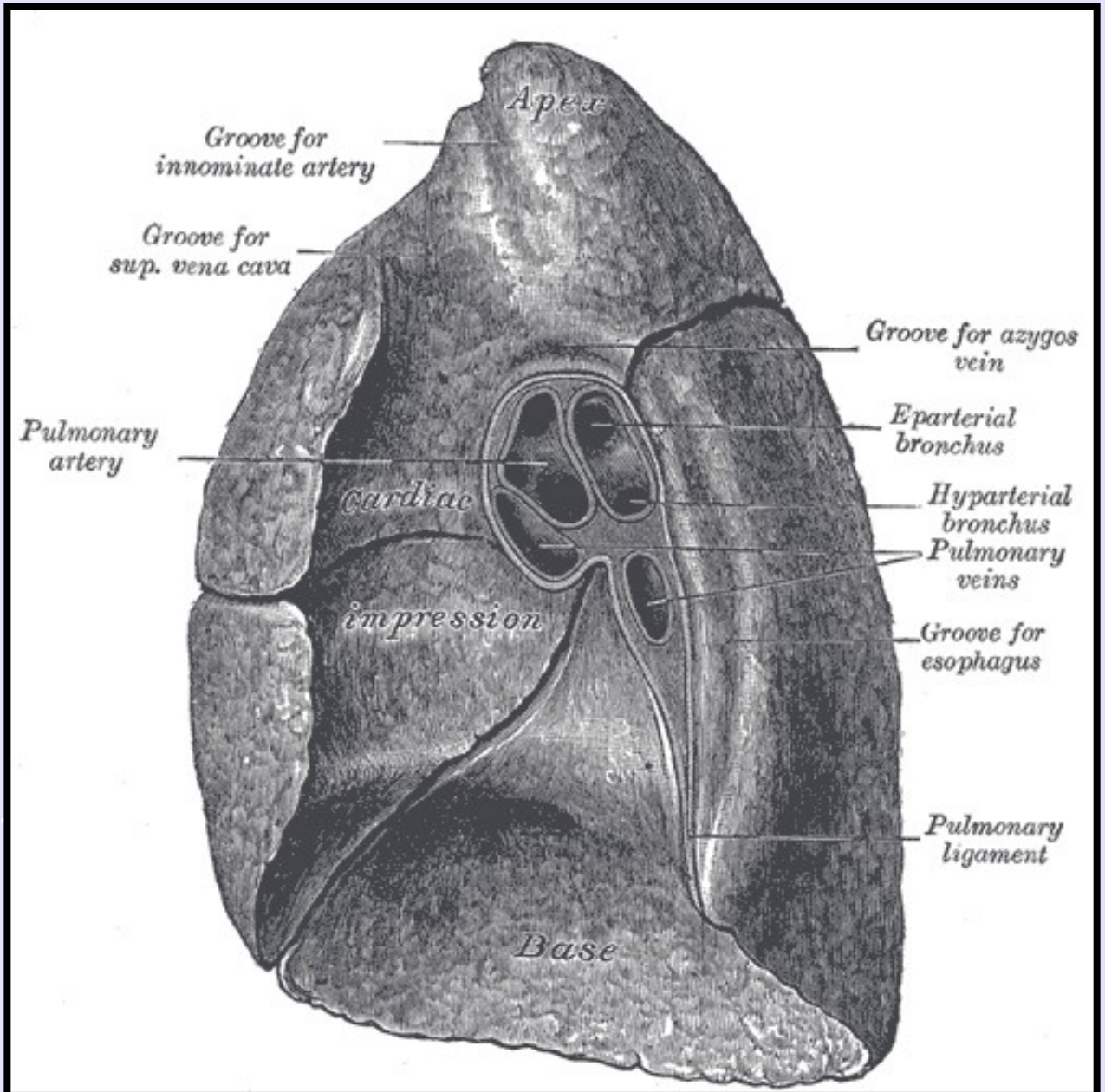

8th thoracic intervertebral disc
Right sympathetic trunk
Costal (parietal) pleura
Hemiazygos vein
Left sympathetic trunk
Neck of rib
Left greater thoracic splanchnic nerve
Mediastinal pleura
Right greater thoracic splanchnic
Azygos vein
Left leaflet of central tendon
Right leaflet of central
Esophagus
Diaphragmatic pleura (cut away)
Inferior vena cava
Thoracic duct
Middle leaflet of central tendon covered by pericardium
Thoracic descending aorta
Diaphragmatic pleura (cut away)
Right phrenic nerve and pericardiacophrenic artery and vein
Mediastinal pleura and pericardium (cut)
Left phrenic nerve and pericardiacophrenic
Right costodiaphragmatic recess
Pericardium
Left costomediastinal recess
Transversus thoracis muscle
Left internal thoracic artery and veins
Right internal thoracic artery and veins
5th costal cartilage
Right costomediastinal recess
Sternum
Anterior mediastinum

MEDIASTINAL PLEURA

- Passes back from the sternum to the vertebral column lining the mediastinum.
- Reflected at the root of the lung becoming the visceral pleura.
- Below the root of the lung the two layers hang down in a deep fold called the pulmonary ligament.


cervical pleura
(parietal pleura)
costal pleura
(parietal pleura)
left lung
oblique fissure
bronchi
upper lobe
visceral pleura
mediastinal pleura
(parietal pleura)
pulmonary veins
cuff of pleura
lower lobe
pulmonary ligament
diaphragmatic pleura
(parietal pleura)

Attachment of Mediastinal pleura showing formation of pulmonary ligament



COSTODIAPHRAGMATIC RECESS

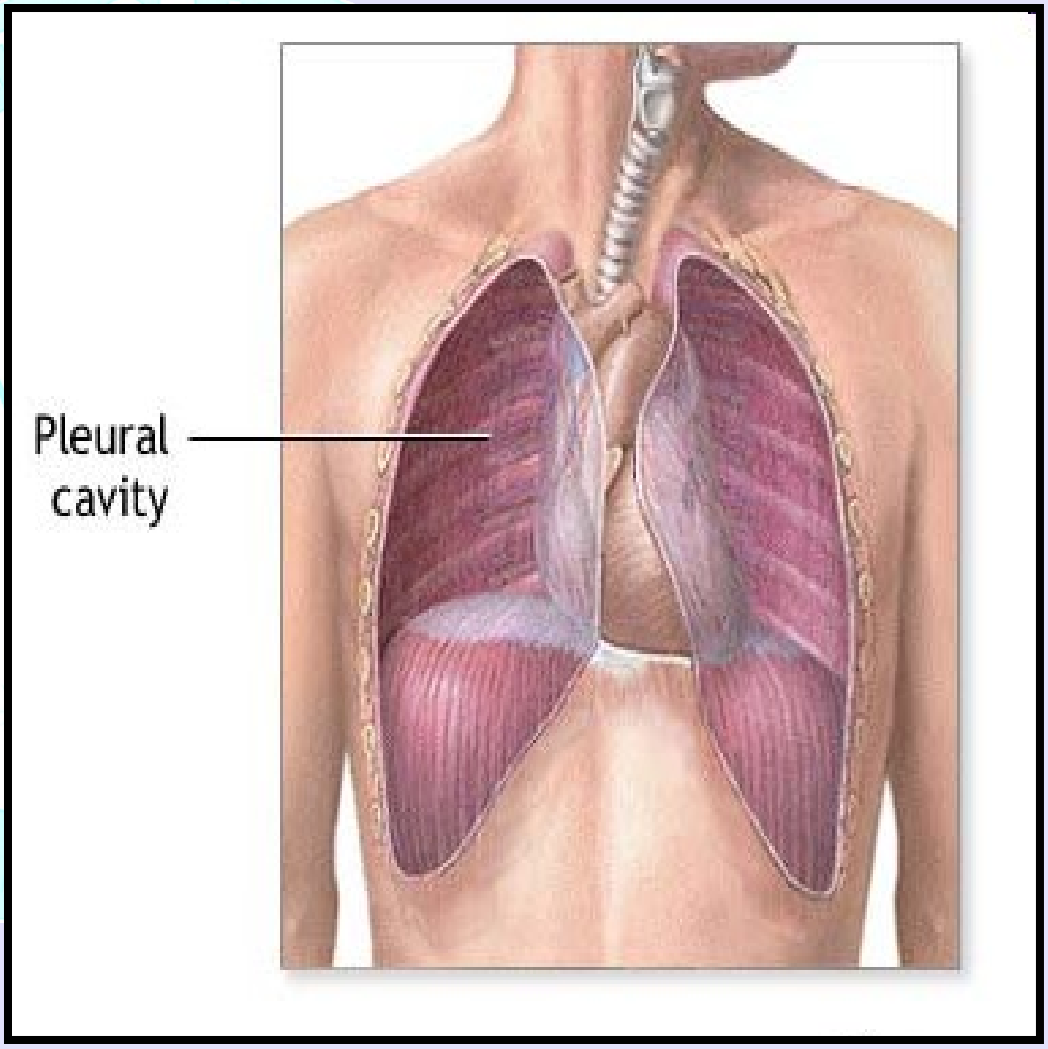
- Are slitlike spaces between the costal and diaphragmatic parietal pleurae
- Separated only by a capillary layer of pleural fluid
- During inspiration, the lower margins of the lungs descend into the recesses
- During expiration, the lower margins of the lungs ascend so that the costal and diaphragmatic pleurae come together again

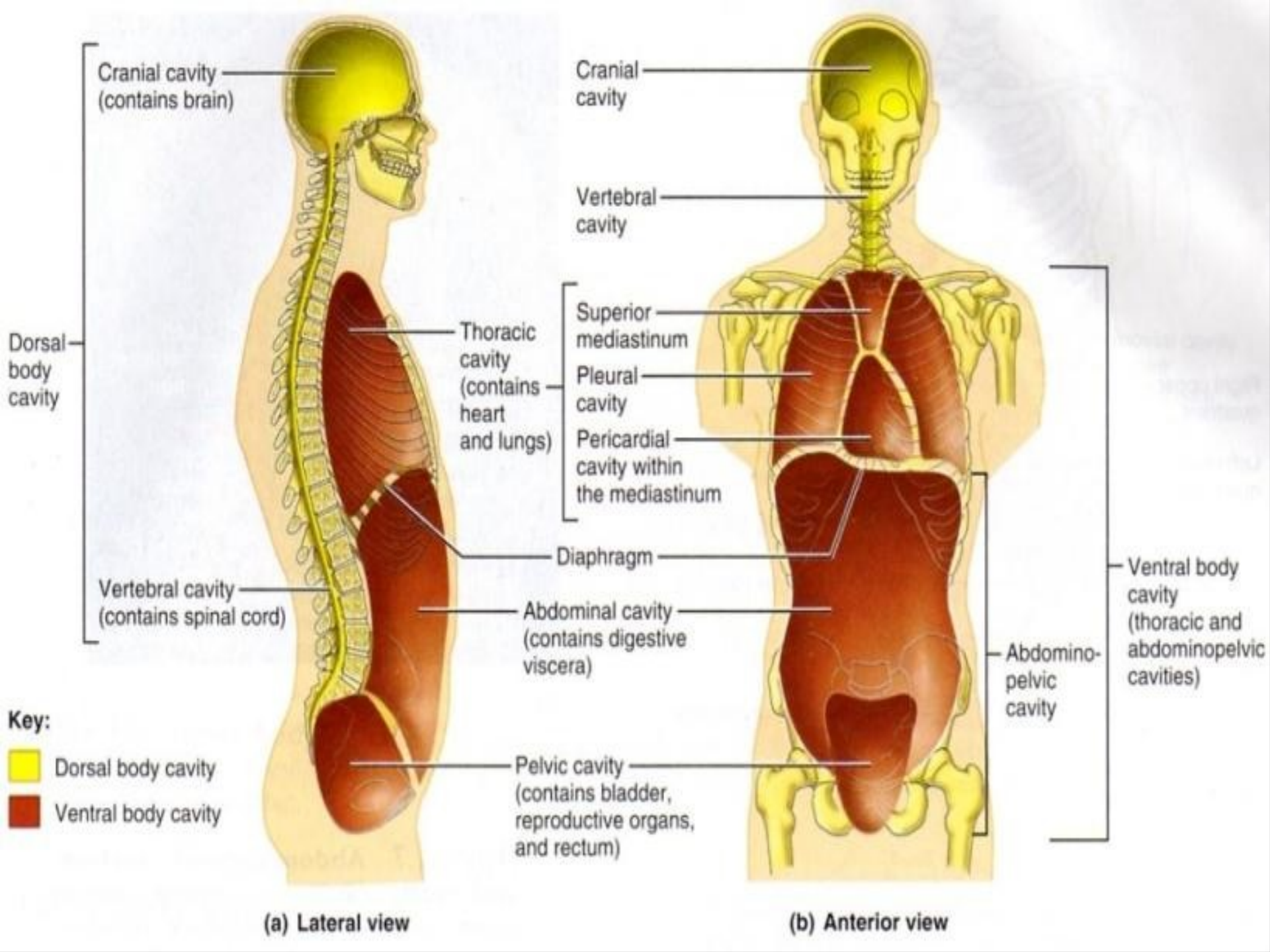
COSTOMEDIASTINAL RECESS

- Are situated along the anterior margins of the pleura
- They are slitlike spaces between the costal and the mediastinal parietal pleurae
- Separated by a capillary layer of pleural fluid
- During inspiration and expiration, the anterior borders of the lungs slide in and out of the recesses

PLEURAL CAVITY

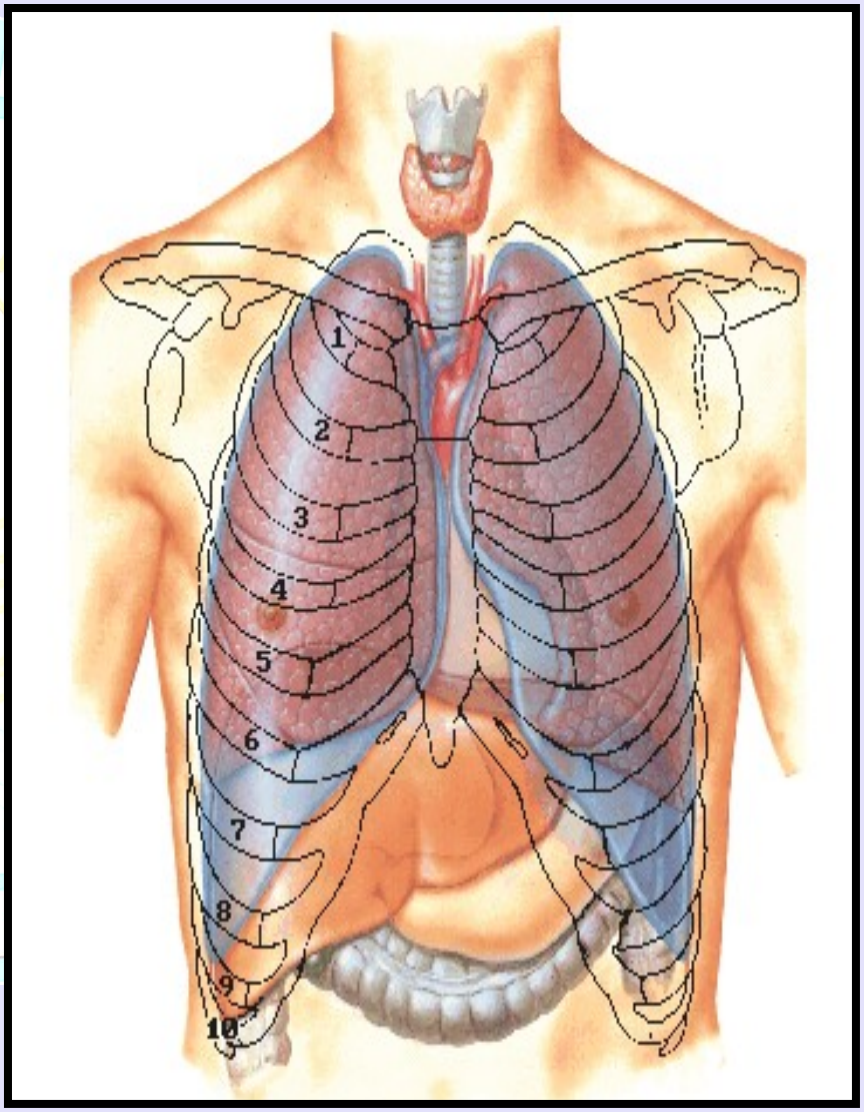
- Potential space between visceral and parietal pleurae
- A thin film of serous fluid lubricates the plaural cavity




Cranial cavity (contains brain)
Dorsal body cavity
Vertebral cavity (contains spinal cord)
Thoracic cavity (contains heart and lungs)
Diaphragm
Abdominal cavity (contains digestive viscera)
Pelvic cavity (contains bladder, reproductive organs, and rectum)
Key:
Dorsal body cavity
Ventral body cavity
(a) Lateral view
Cranial cavity
Vertebral cavity
Superior mediastinum
Pleural cavity
Pericardial cavity within the mediastinum
Abdomino-pelvic cavity
Ventral body cavity (thoracic and abdominopelvic cavities)
(b) Anterior view

SURFACE ANATOMY-RIGHT PLEURA

- Anterior, down the middle line of the sternum along the seventh costal cartilage, and across the bony extremities of the eighth and ninth ribs.
- In the mid-axillary line the pleura reaches its-lowest limit- that of the bony tip of the tenth rib. It then ascends slightly, cutting across the eleventh rib to the middle of the twelfth rib and on to the first lumbar vertebra.
- The posterior line of reflection is to the left of the mid-line of the vertebral bodies, the aorta intervening between them.



SURFACE ANATOMY-LEFT PLEURA

- This only differs from the right in the anterior line of reflection.
- Opposite the fourth costal cartilage the left pleura deviates to the left, leaving a small area of pericardium uncovered.
- It cuts across the fifth, sixth, and seventh costal cartilages to the bony tip of the eighth rib, after which the line-of reflection resembles that of the right pleura.

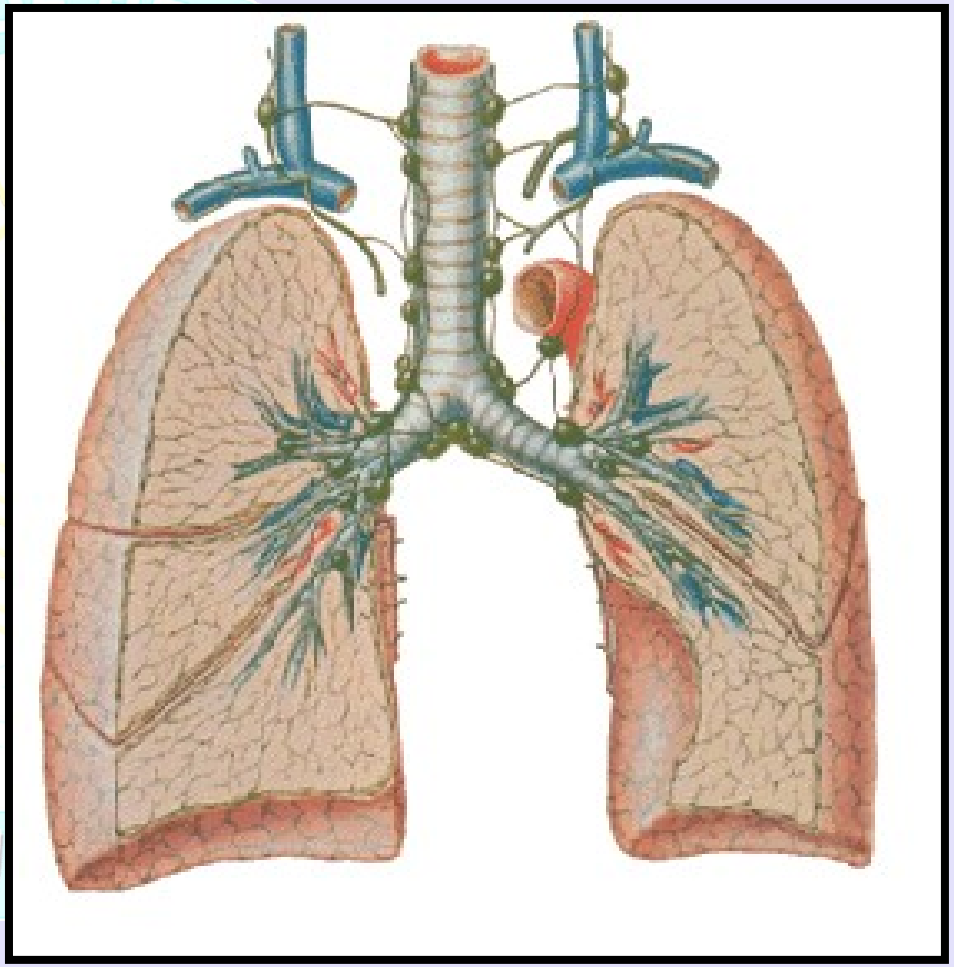
VASCULAR SUPPLY

- Visceral pleura:
  - Bronchial and pulmonary arterial system
  - Venous drainage-pulmonary vein
- Parietal pleura
  - Systemic vessels supplying the chest wall, diaphragm and mediastinal structures
  - Venous drainage-Superior vena cava

LYMPHATIC DRAINAGE

- Sub pleural space of the visceral pleura has large network of lymphatic channel
- Visceral pleura: pulmonary plexus located in interlobar and peribronchial space.
- Parietal pleura:

  parietal pleural lymphatics into the internal mammary and intercostal chains



NERVE SUPPLY

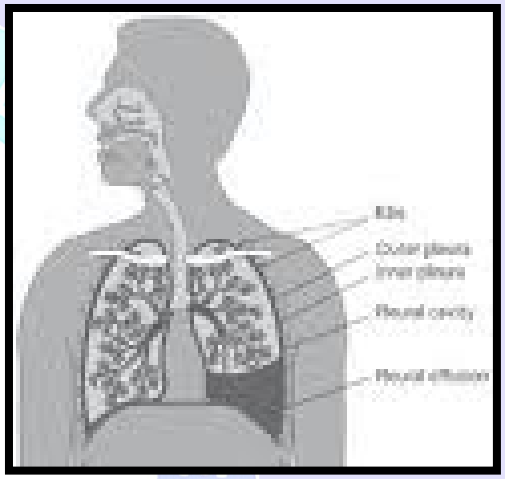
- The parietal pleura is sensitive to pain, temperature, touch and pressure, and is supplied as follows:
- The costal pleura is segmentally supplied by the intercostal nerves
- The mediastinal pleura is supplied by the phrenic nerve
- The diaphragmatic pleura is supplied over the domes by the phrenic nerve and around the periphery by the lower six intercostal nerves

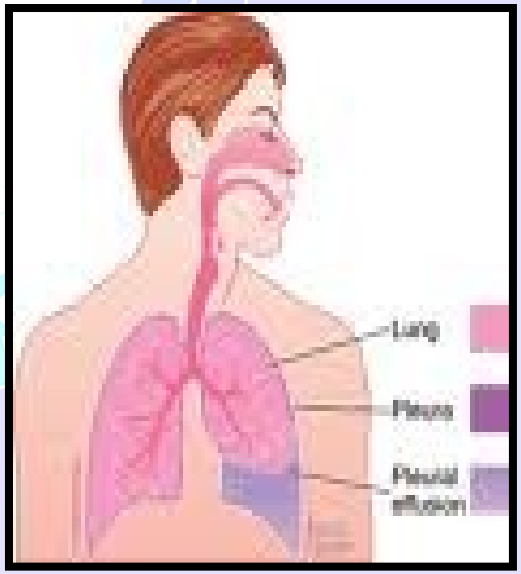
NERVE SUPPLY

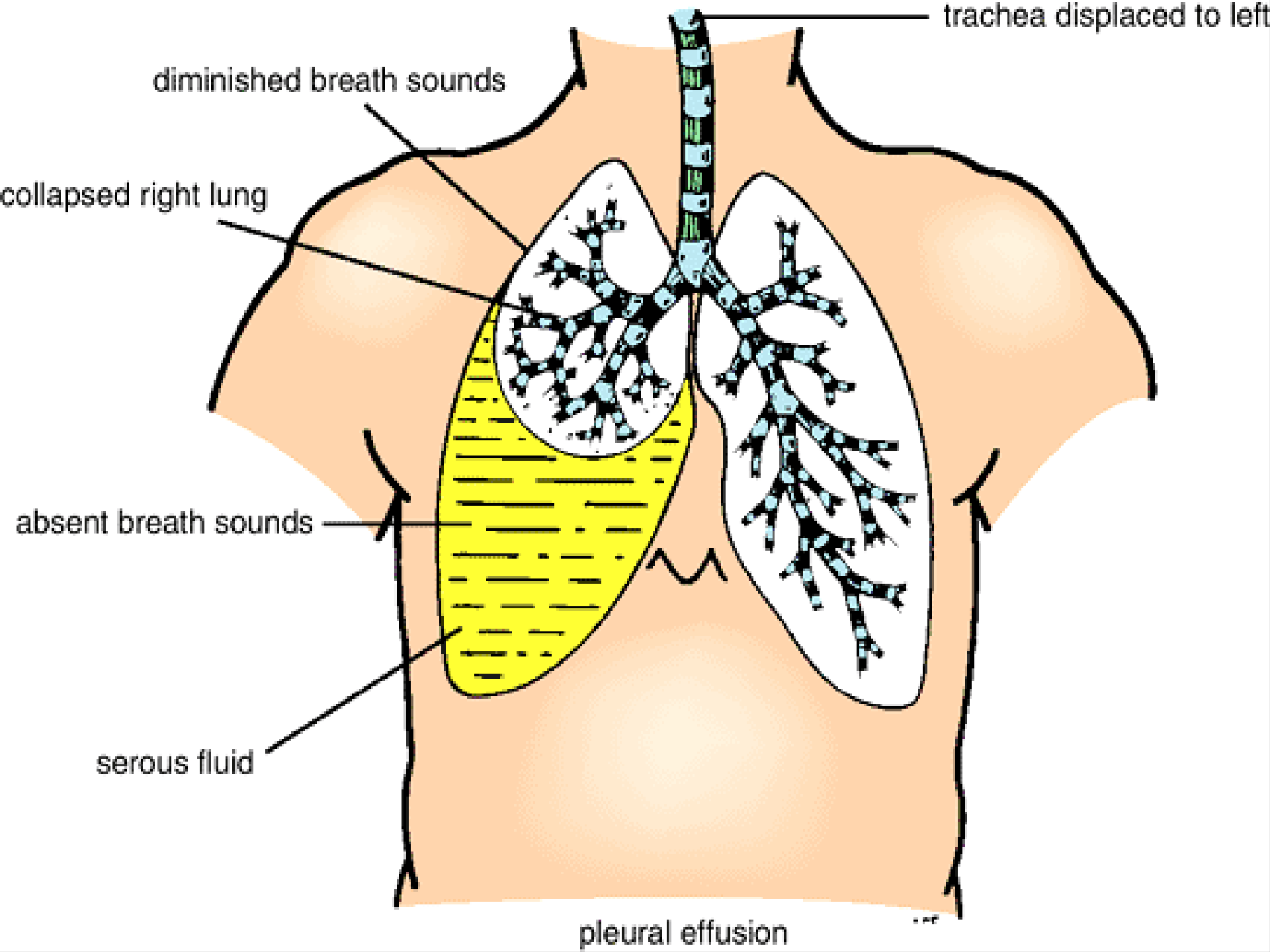
- The visceral pleura covering the lungs is sensitive to stretch
- It is insensitive to common sensations such as pain and touch
- It receives an autonomic nerve supply from the pulmonary plexus

PLEURAL EFFUSION

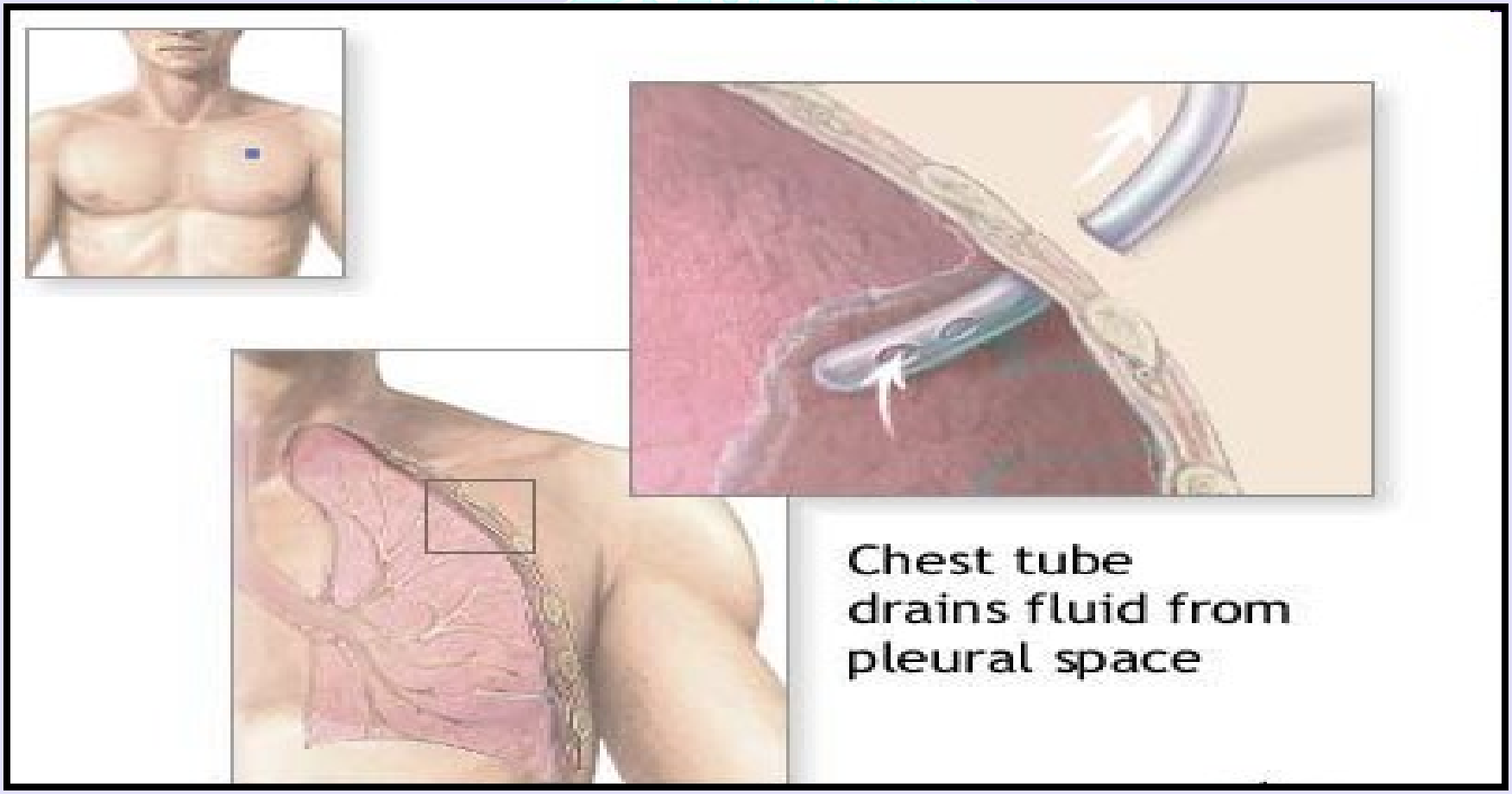
- Presence of fluid in Pleural space
- Can be transudative or exudative
- Transudate (low protein content)
  - cardiac failure
  - Hypoproteinaemia
- Exudate (high protein content)
  - pneumonia
  - TB
  - connective tissue disease
  - malignancy (primary or metastatic)






trachea displaced to left
diminished breath sounds
collapsed right lung
absent breath sounds
serous fluid
pleural effusion

CHEST TUBE DRAINAGE



PNEUMOTHORAX

Presence of gas (air) in pleural space

- Simple spontaneous pneumothorax
- Tension pneumothorax

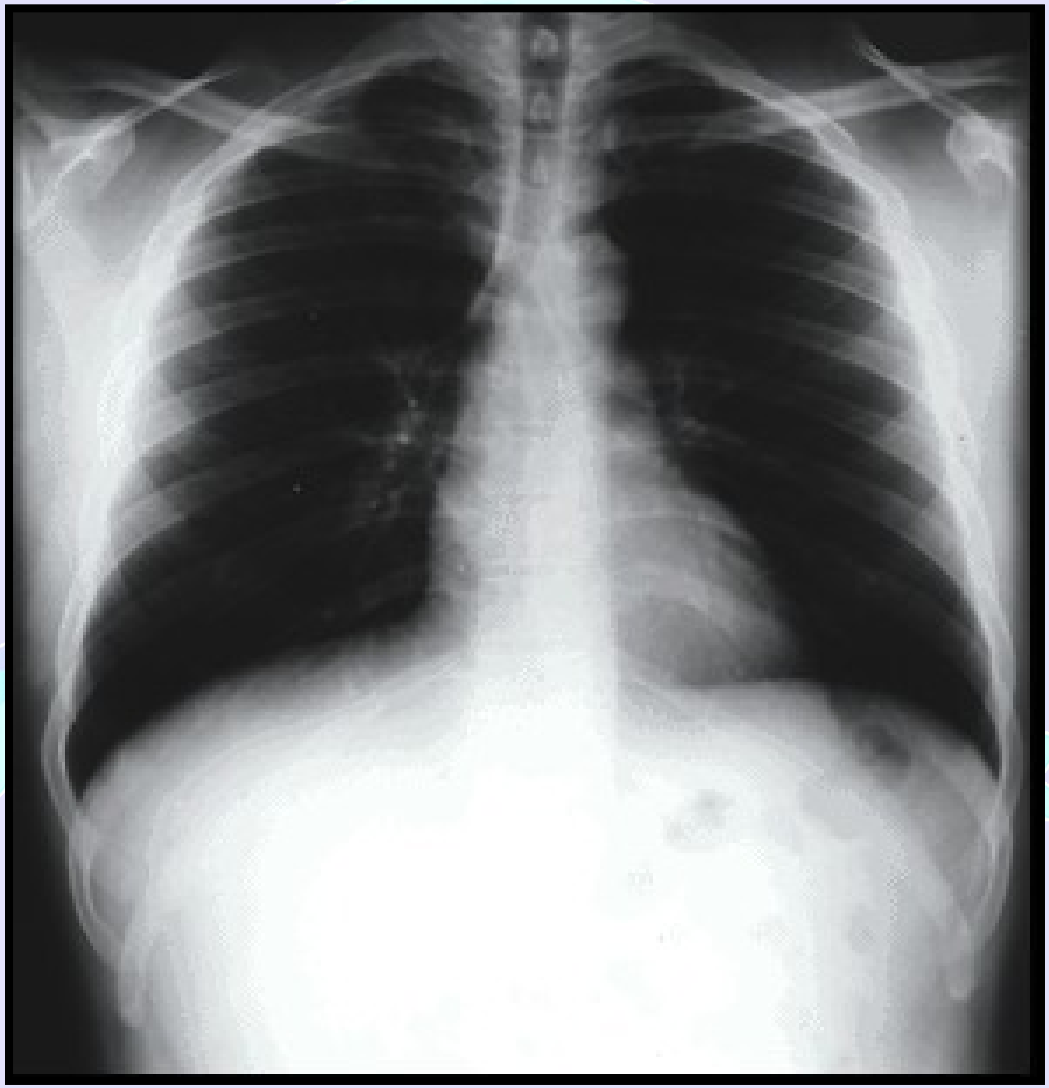
SPONTANEOUS PNEUMOTHORAX

- Spontaneous pneumothorax is rarely seen before puberty. Children with spontaneous pneumothorax often have underlying disease such as cystic fibrosis.
- More common in men (6:1).
- More common in smoker.
- Typical patient is young, tall, thin man in late adolescence and early adulthood.

TENSION PNEUMOTHORAX

- **Clinical presentation**: sudden onset of severe shortness of breath.
- **Symptoms and sign**: tachycardia, sweating, hypotension, mediastinal shift.
- **Clinical diagnosis** of tension pneumothorax made on basis of appropriate history and physical findings.
- **Emergency placement** of chest tube without confirmatory chest radiography.

Normal Chest Radiograph
PA View

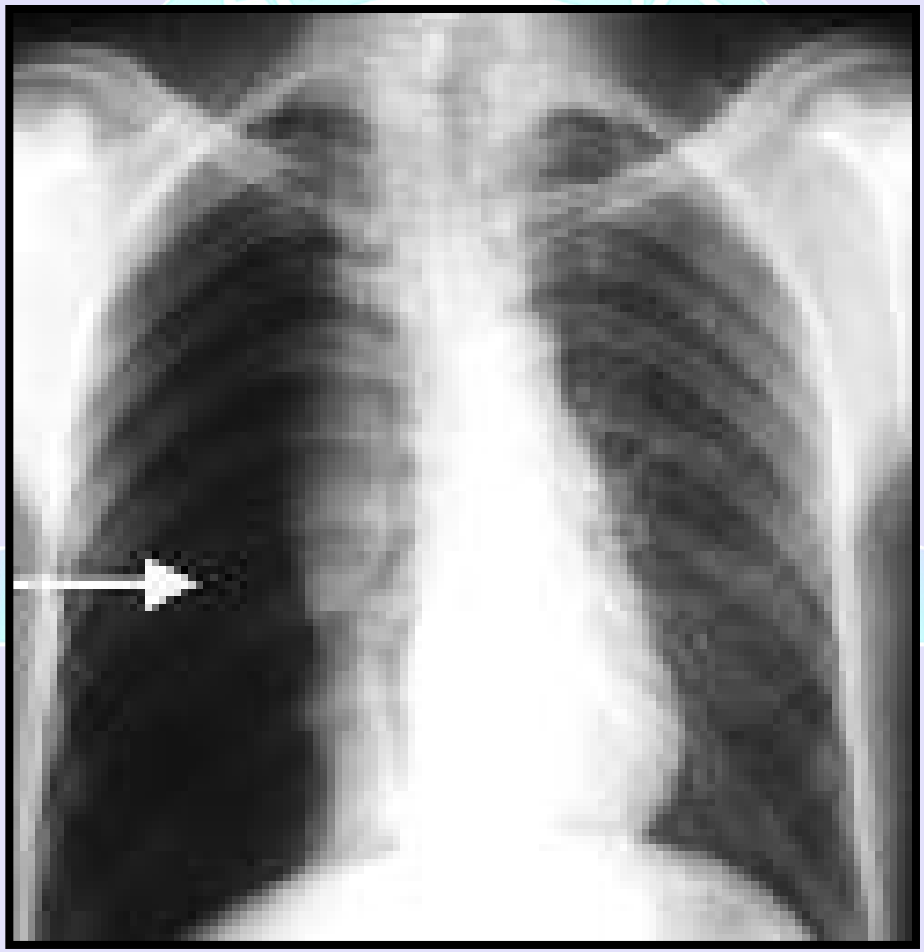


Pleural effusion-left lung





X-RAY PNEUMOTHORAX



REFERENCES

- **CLINICAL ORIENTED ANATOMY BY K.L.M**

6TH EDITION

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THANK YOU